

Dept. of Electronics and Electrical Communication Engineering
Indian Institute of Technology Kharagpur



VLSI Design Laboratory (EC69216)

Prasit Mazumder (22EC37029)

Group Number: 06

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Experiment 5: Schematic and Physical Design of a 4-bit Adder and a Pipeline Unit

1 Objective

- Design and simulate a 1-bit Full Adder circuit and measure its critical path delays.
- Perform layout design of the Full Adder and verify using DRC and LVS.
- Compare pre-layout and post-layout performance.
- Design a 4-bit Ripple Carry Adder (RCA) using the 1-bit Full Adder.
- Construct a 4-bit Parallel-In Parallel-Out (PIPO) register using D Flip-Flops.
- Design and simulate a pipelined adder using register stages.

2 Theory

A Full Adder is a combinational logic circuit that performs addition of three input bits: A , B , and C_{in} . It has two outputs: Sum and Carry-out.

$$Sum = A \oplus B \oplus C_{in}$$

$$C_{out} = AB + BC_{in} + AC_{in}$$

A 4-bit Ripple Carry Adder (RCA) is built by connecting four 1-bit Full Adders, where the carry output of one stage drives the carry input of the next stage.

In pipelined designs, registers are placed between the combinational logic modules to increase the system throughput by allowing multiple operations to overlap in time.

3 Part (a): Design and Simulation of 1-bit Full Adder

3.1 NAND Gate Schematic

The full adder circuit is implemented using NAND gates. The schematic of the NAND gate used as the basic building block is shown below.

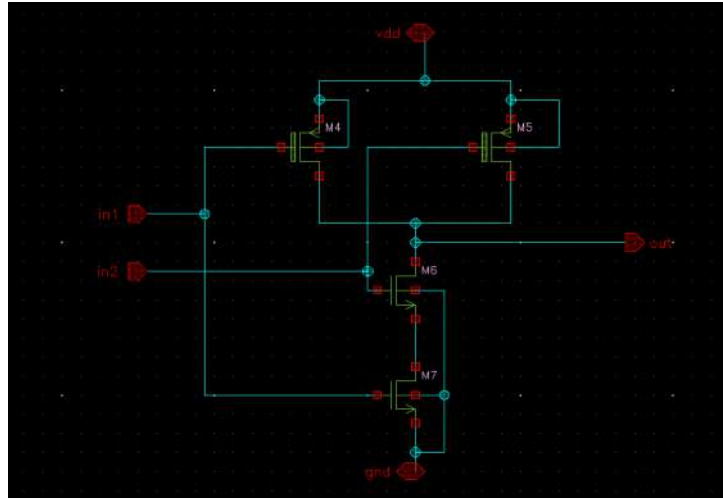


Figure 1: Schematic of NAND gate

3.2 Full Adder Schematic

The 1-bit full adder was designed using NAND gates. The schematic implementation is shown below.

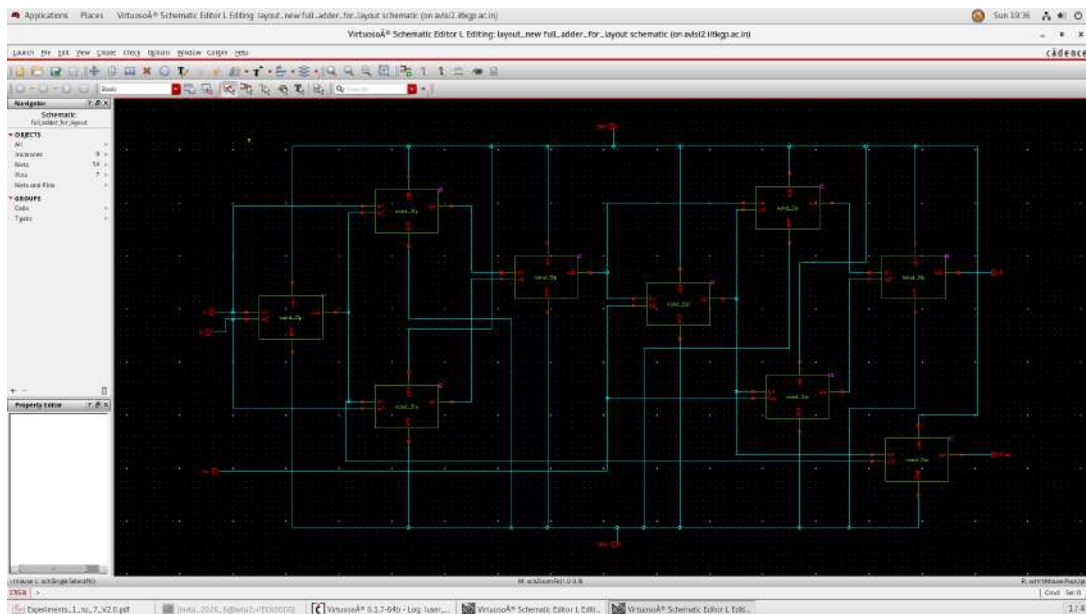


Figure 2: Schematic of 1-bit Full Adder

3.3 Testbench

A testbench was created to apply different combinations of inputs A , B , and C_{in} to verify the functionality of the full adder.

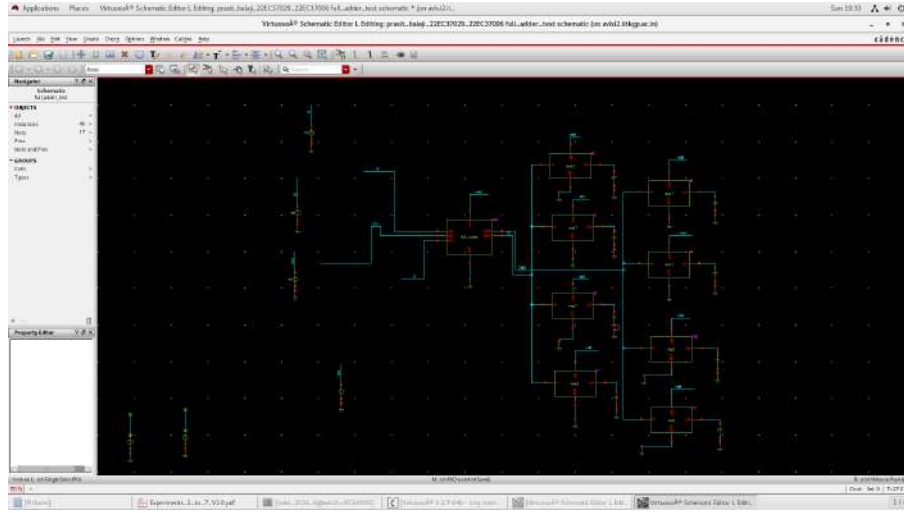


Figure 3: Testbench used for Full Adder simulation

3.4 Simulation Waveform



Figure 4: Simulation waveform showing Sum and Carry outputs

3.5 Delay Measurement

Propagation delays were measured from input transitions to output transitions.

- Best case delay (Input $\rightarrow$ Sum): 135.107 ps
- Worst case delay (Input $\rightarrow$ Sum): 237.218 ps
- Best case delay (Input $\rightarrow C_{out}$): 82.516 ps
- Worst case delay (Input $\rightarrow C_{out}$): 94.032 ps

The load capacitance was set equal to the input capacitance of four unit-size inverters.

4 Part (b): Physical Design of Full Adder

4.1 NAND Gate Layout

The layout of the NAND gate used in the full adder design is shown below.

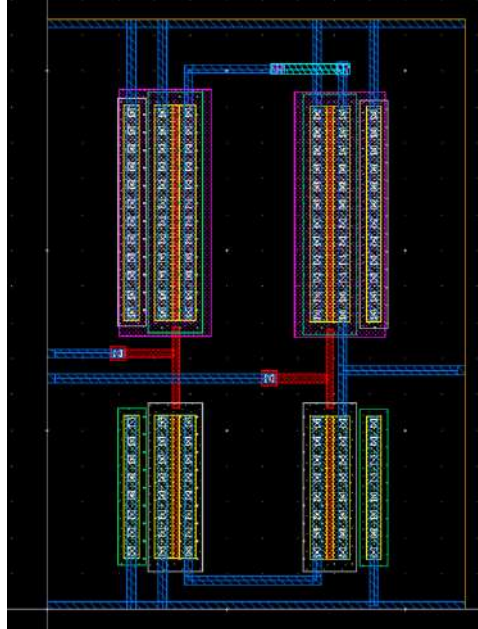


Figure 5: Layout of NAND gate

4.2 Full Adder Layout

The physical layout of the 1-bit full adder was implemented using standard CMOS design rules.

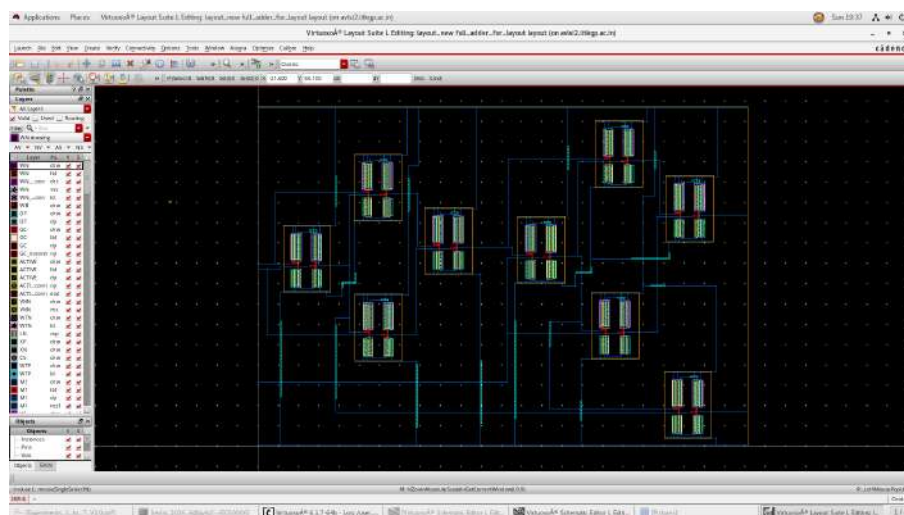


Figure 6: Layout of the Full Adder

4.3 DRC Verification

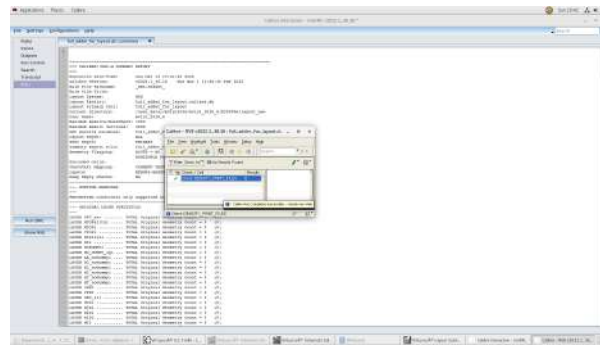


Figure 7: DRC verification result

The layout passed all design rule checks successfully.

4.4 LVS Verification

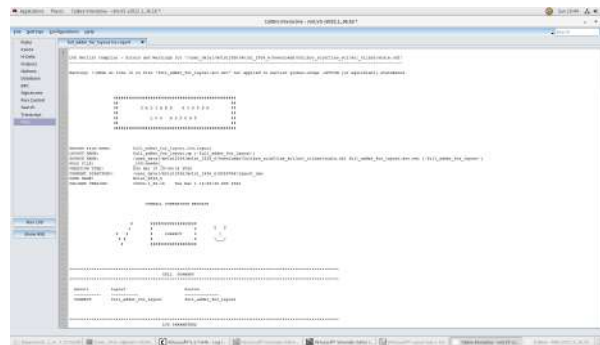


Figure 8: LVS verification result

The LVS check confirms that the layout matches the schematic.

4.5 Parasitic Extraction

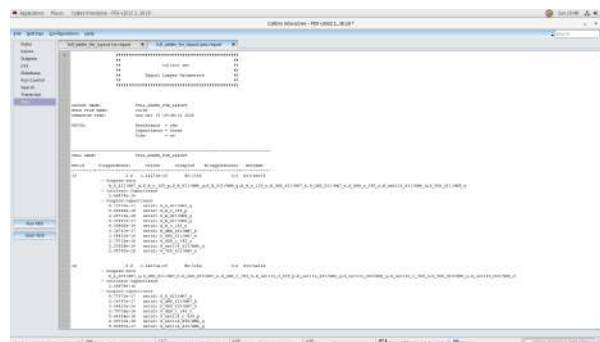


Figure 9: Part of PEX report showing extracted parasitics

4.6 Post-layout Waveform



Figure 10: Post-layout simulation waveform

4.7 Post-layout Delay

- Best case delay (Input to Sum): 162.250 ps
- Worst case delay (Input to Sum): 280.057 ps
- Best case delay (Input to C_{out}): 102.616 ps
- Worst case delay (Input to C_{out}): 119.709 ps

5 Part (c): Design of 4-bit Ripple Carry Adder

5.1 Schematic

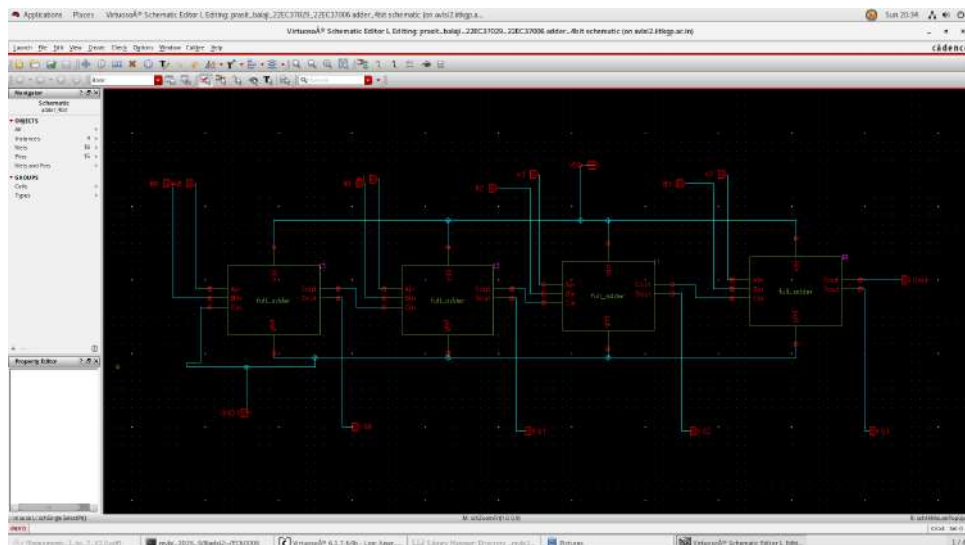


Figure 11: Schematic of 4-bit Ripple Carry Adder

5.2 Waveform



Figure 12: Simulation waveform of 4-bit Ripple Carry Adder

The simulation verifies correct addition for different input combinations.

6 Part (d): Design of 4-bit PIPO Register

6.1 D Flip-Flop

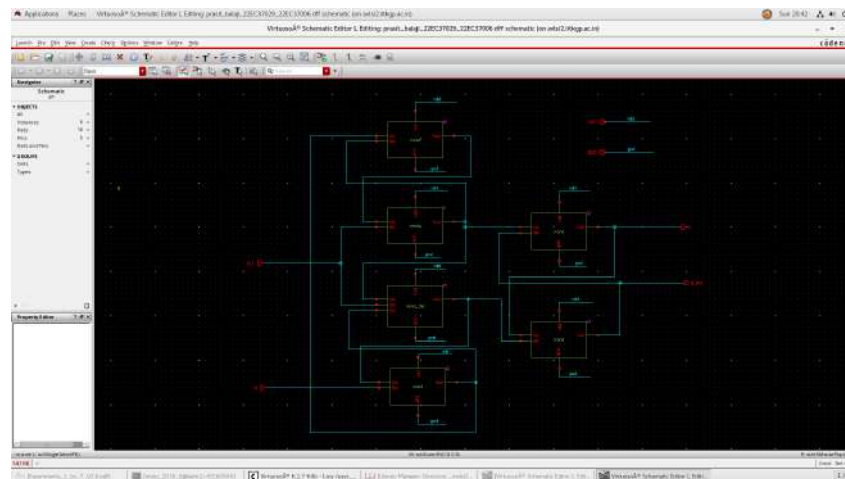


Figure 13: Schematic of D Flip-Flop

6.2 4-bit PIPO Register

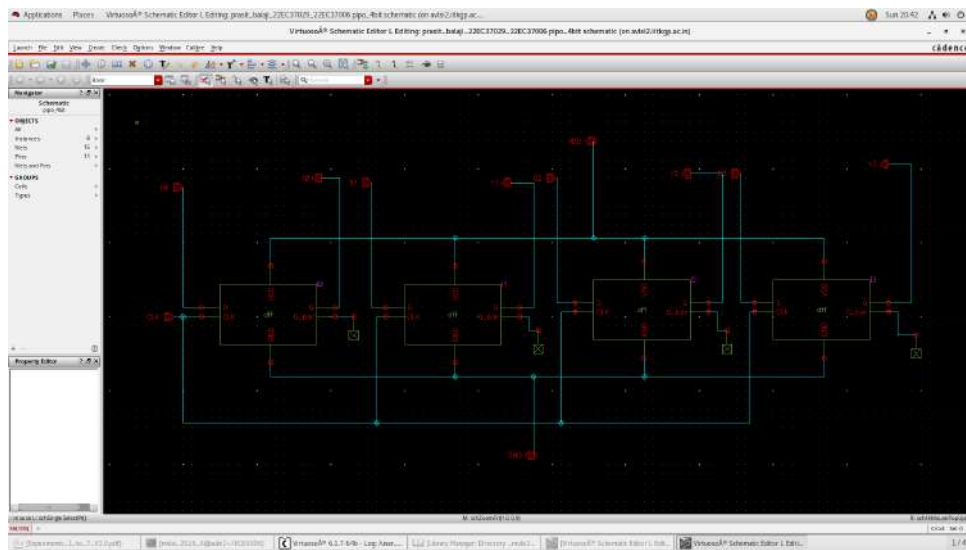


Figure 14: Schematic of 4-bit PIPO Register

The register stores a 4-bit input value and transfers it to the output on the active clock edge.

7 Part (e): Pipelined Adder

7.1 Schematic

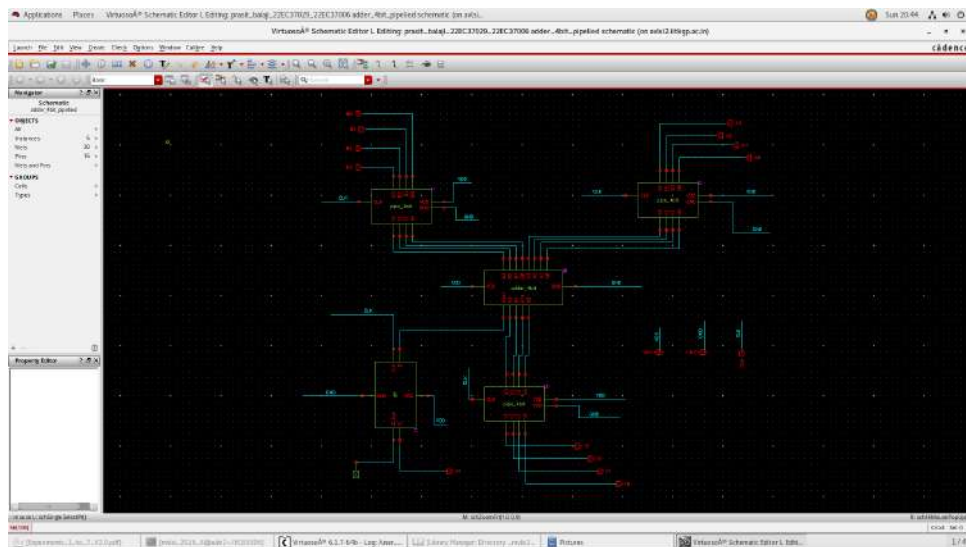


Figure 15: Pipelined 4-bit Adder

7.2 Waveform

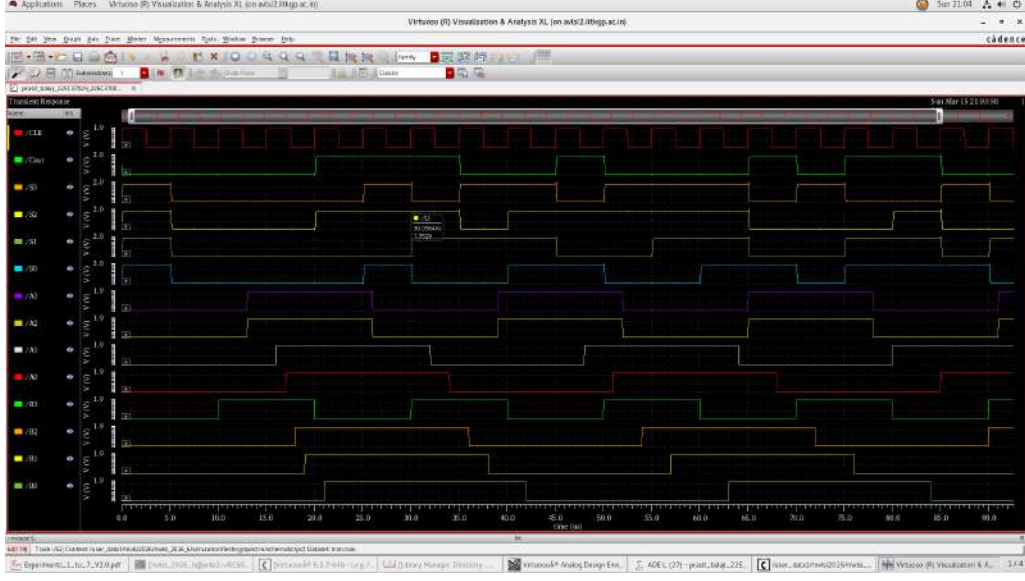


Figure 16: Waveform of pipelined adder

7.3 Maximum Operating Frequency

The maximum operating frequency of the pipeline is determined from the minimum clock period.

$$T_{clk_{min}} = 1.2 \text{ ns}$$

$$f_{max} = \frac{1}{T_{clk_{min}}} = \frac{1}{1.2 \times 10^{-9}}$$

$$f_{max} = 833.33 \text{ MHz}$$

8 Discussion

- The pre-layout simulation of the entire adder circuit confirms the logical validity of the designed circuit and also gives an estimate of the initial propagation delay of the circuit. The schematic-level simulation does not take into consideration any parasitic effect of the circuit layout or routing and hence represents ideal circuit behavior.
- The post-layout simulation shows that the propagation delay is longer than it was in the pre-layout simulation. The propagation delay increases because the layout adds parasitic capacitances and resistances that are linked to metal interconnects, diffusion regions, and the closeness of devices.
- The layout design successfully passed the Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification. The DRC verifies whether the layout design adheres to all the fabrication rules,

while LVS verifies whether the layout design matches the schematic design in terms of electrical connections.

- The ripple carry adder also shows the concept of sequential carry signals propagating from the least significant bit to the most significant bit in the addition process. It has been observed that the total delay in the addition process increases linearly with the number of bits in the adder.
- The addition of pipeline registers enhances the overall efficiency of the system by allowing multiple stages in the computation to be executed simultaneously. It has been observed that the overall delay in the addition process does not change, but the efficiency of the overall system enhances considerably.
- It has been observed that the overall frequency of the pipelined adder is limited to around 833.33 MHz, which is the maximum frequency allowed in the addition process before the signals do not get enough time to settle before the next clock edge, which can lead to incorrect signals being captured in the register stages.

9 Conclusion

In this experiment, a 1-bit full adder was designed and simulated, and its physical layout was implemented and verified. The pre-layout and post-layout performances were compared to examine the effect of parasitics. A 4-bit ripple carry adder was built using the designed full adder, and a PIPO register was implemented using D flip-flops. Finally, a pipelined adder architecture was designed and simulated to determine the maximum operating speed of the system.